Aleksei Lian

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US: authorized to work, no sponsorship needed now or in the future | Canada: eligible to work full-time in Summer 2027

Education

York University — Toronto, ON
Bachelor of Science in Computer Science — *Expected Dec. 2027*
Transferred from Carleton University — *Sep. 2023 – Dec. 2024*

Experience

Software Engineer (Co-Founder) — May 2026 – Present
Arzan (China-to-Kyrgyzstan E-Commerce, Pre-Launch) — *Remote*

- Built a 90+ endpoint FastAPI/PostgreSQL API, React staff console, and React Native app as the sole engineer.
- Fixed a race where two operators could claim one order and have it bought twice, using PostgreSQL row locks.
- Built the job (not live yet) that hands paid orders to our buying agent: it claims each with FOR UPDATE SKIP LOCKED and commits the claim before the call, so a crash leaves a row to reconcile, not a second purchase.
- Made checkout idempotent with a unique key on user and attempt ID, so a retry gets the first order back.
- Enforced allowed money, quantity, and status values with 50 PostgreSQL CHECK constraints.
- Set up GitHub Actions CI that runs the 2,300+ test backend suite on PostgreSQL and SQLite, plus an alembic check that caught a real model/migration mismatch.

Backend Engineer (Co-Founder) — Apr. 2026 – Aug. 2026
Tanish (Dating App for Central Asia, iOS Beta) — *Remote*

- Built an iOS app (TestFlight beta, 2 testers): FastAPI/PostgreSQL backend, WebSocket chat, React Native client.
- Designed JWT auth with 15-minute access tokens and rotating refresh tokens that revoke the chain on reuse.
- Added AWS Rekognition face-liveness checks, using STS credentials that can only start the liveness stream.
- Found that anyone with the project's public key could read and write user tables through Supabase's auto-generated API, and enabled deny-by-default row-level security on them.
- Fixed moderation jobs that silently never retried after an OpenAI quota outage: added real retries and an hourly sweep that re-screens missed chat and voice content.

Projects

Temir AI - AI Quoting Startup (Co-Founder) | *JavaScript, Anthropic API, PostgreSQL* — Aug. 2026 – Present

- Built the product for our pilot customer, a metal supplier, where **4 sales managers at 2 warehouses** made about 70 quotes with it from August to October, from orders sent as chat text, PDF, spreadsheet, or photo.
- Kept prices out of the LLM's output: it maps each line to a catalog ID, prices come from the catalog, and inexact matches are flagged for a manager.
- Capped API spend per user and company-wide by reserving budget under a PostgreSQL advisory lock.

Mendy - AI Advice Companion | *Python, FastAPI, PostgreSQL, OpenAI API* — Jul. 2026

- Built the backend for an AI advice companion with long-term memory, proactive follow-ups, and crisis escalation.
- Caught a prompt edit that lowered a crisis message's risk tier, using a 40-message safety test set that must flag all 12 crisis cases and is re-run by hand after every safety-prompt change.

URL Shortener | *Python, FastAPI, asyncpg, PostgreSQL, Docker* — Feb. – Mar. 2026, Nov. 2026

- Built an async URL shortener where the primary key rejects a duplicate short code and the insert retries with a new one, so a collision doesn't fail the request.
- GitHub Actions CI runs 83 tests against live PostgreSQL, including forced-collision and concurrency tests; an earlier Flask version ran on Cloud Run.

Chess Engine | *Python, UCI* — Feb. 2026, Nov. 2026

- Built a UCI chess engine that plays timed games, using alpha-beta search and a transposition table.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Java, C
Frameworks: FastAPI, SQLAlchemy, Pydantic, Alembic, pytest, React, React Native, LLM APIs (OpenAI, Anthropic)
Data & Cloud: PostgreSQL (transactions, locking), Redis, SQLite, AWS, GCP (Cloud Run), OpenTofu
Developer Tools: Git, GitHub Actions (CI/CD), Docker, Linux, REST APIs, Claude Code, Codex, Antigravity